

L1.1b Transformations · §1.3

Outside or inside — what it moves, and which way

THE RULE

$$y = \underbrace{a}_{\text{R}} f\left(\underbrace{b}_{\text{D}} \left(x + \underbrace{c}_{\text{D}}\right)\right) + \underbrace{d}_{\text{R}}$$

“Anything outside the $f(x)$ affects RANGE normally. Anything inside the $f(x)$ affects DOMAIN backward.”

OUTSIDE | INSIDE

OUTSIDE the $f(x)$

$\Rightarrow$ moves **RANGE**, normally

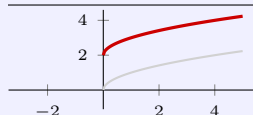
$f(x) + d$ slide d in the $+y$ direction

$a f(x)$, $|a| > 1$ stretch vertically

$a f(x)$, $0 < |a| < 1$ compress vertically

$-f(x)$ reflect over the x -axis

$|f(x)|$ flip whatever is below the x -axis up over it



$\sqrt{x} \rightarrow \sqrt{x} + 2$: R climbs to $[2, \infty)$, D never moves.

INSIDE the $f(x)$

$\Rightarrow$ moves **DOMAIN**, backward

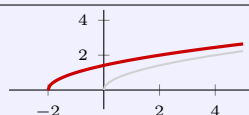
$f(x + c)$ slide c in the $-x$ direction †

$f(bx)$, $|b| > 1$ **compress** horizontally †

$f(bx)$, $0 < |b| < 1$ **stretch** horizontally

$f(-x)$ reflect over the y -axis

$f(|x|)$ chop off the negative x -values and reflect the positive values back over the y -axis

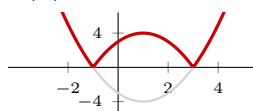


$\sqrt{x} \rightarrow \sqrt{x + 2}$: D slides to $[-2, \infty)$ — the $+2$ went the *other* way.

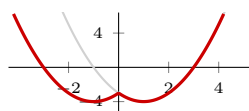
† The two rows that read wrong are the whole point. **Never say “shift left”** — say the direction.

THE ABSOLUTE VALUE PAIR — same bars, opposite sides

$f(x) = x^2 - 2x - 3$ — Loh: $m = 1$, $d = 2$, roots 1 ± 2 , so zeros at -1 and 3 and vertex $(1, -4)$.



$|f(x)|$ — **outside**. $R = \{y \mid y \geq 0\} = [0, \infty)$



$f(|x|)$ — **inside**. $R = \{y \mid y \geq -4\} = [-4, \infty)$

“outside bars bend the outputs up; inside bars chop and mirror the inputs”

FIVE COMBOS — A is f 's domain, B is g 's

$$f \pm g, \quad fg: \quad A \cap B \qquad \frac{f}{g}: \quad \{x \in A \cap B \mid g(x) \neq 0\}$$

“Intersection cuts out just the overlap of the two domain intervals; union gives you any place that either of them goes.”

Complete the square to read a parabola's shifts: $x^2 + 6x + 10 = (x + 3)^2 + 1$, vertex $(-3, 1)$ — slide x^2 3 in the $-x$ direction and 1 in the $+y$ direction.

COMPOSITION — diffnifod

$$\text{Domain}(f(g(x))) = D_{If} \cap I_{f \rightarrow OD}$$

$x \rightarrow \boxed{g} \rightarrow g(x) \rightarrow \boxed{f} \rightarrow f(g(x))$ D_{If} — the **inner** function has to be defined.
 $\quad \quad \quad \downarrow \quad \quad \quad \downarrow$ $I_{f \rightarrow OD}$ — the outer has to be defined too, but its input is $g(x)$:
 $\quad \quad \quad D_{If} \quad \quad \quad I_{f \rightarrow OD}$ push the outer domain **backward** through the inner.

“Domain of Inner Function, intersect Inner Function to Outer Domain — diffnifod”

Roles, not letters. In $g(f(x))$ the inner function is f , so D_{If} means f 's domain.

Four shapes come out: an interval $[-\frac{1}{5}, 10]$ · a union $(-\infty, -3] \cup [3, \infty)$ · a hole punched in an interval $[0, 4) \cup (4, \infty)$ · **empty** $\emptyset$.

WATCH OUT

✗ $f(x + 2)$ shifts the graph 2 to the right → **inside, so backward** — slide 2 in the $-x$ direction

✗ $f(3x)$ stretches it horizontally by 3 → **inside, so backward** — $|b| > 1$ *compresses* by 3

✗ $|f(x)|$ and $f(|x|)$ do the same thing → **outside bars change outputs, inside bars change inputs**

✗ simplify first, then read the domain off the answer → $(\sqrt{x})^2 - 9 = x - 9$ looks defined everywhere — the domain still remembers $[0, \infty)$

✗ the composite's domain is the outer function's domain → **intersect both halves** — D_{If} and $I_{f \rightarrow OD}$

✗ every composite has a domain → $\sqrt{-9 - x^2}$ — no overlap, nothing to cut out, so $\emptyset$